

AMR SALAMA

Data Scientist

Egypt – New Cairo | amrkhaledsalama16@gmail.com |
+20102349723|github.com/amrk216 |
[linkedin.com/in/amrkhaledsalama/](https://www.linkedin.com/in/amrkhaledsalama/)

Summary

I am a Data Scientist with expertise in Python, SQL, and data visualization, delivering actionable insights through machine learning, deep learning, and natural language processing (NLP). I have successfully developed predictive models and text-processing solutions that building machine learning and NLP solutions to support data-driven decision-making. I excel at transforming complex datasets into clear, insightful visualizations that effectively communicate findings to stakeholders.

Technical Skills

- **Programming:** Python, SQL
- **Databases:** MySQL, MongoDB
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras
- **Natural Language Processing:** NLTK, SpaCy, Hugging Face Transformers, Sentiment Analysis, Named Entity Recognition
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, AWS (Basic)

Projects

Log Classification System

- Built an NLP-based log classification system using spaCy and Scikit-learn, achieving 90% anomaly detection Performance and reducing system downtime by 20%.

Fake News Detection (NLP)

- Fine-tuned a BERT model using Hugging Face Transformers achieving 85% classification Model Performance

Sunflower Leaf Disease Detection

- Developed a CNN model with TensorFlow and OpenCV
- achieving 88% accuracy in early disease detection, reducing crop loss by 10%.

Mini RAG Application

- Built a Retrieval-Augmented Generation (RAG) application for document-based question answering.
- Implemented document parsing, semantic chunking, embedding generation, and vector search using Qdrant.
- Designed FastAPI endpoints and Dockerized the application for scalable deployment.
- Improved retrieval accuracy through optimized embedding and indexing strategies.

Gradient Descent Neural Network from Scratch

- Implemented a custom neural network in Python to gain a deeper understanding of

optimization algorithms and backpropagation.

Education

Bachelor's Degree in Computer Information Systems

New Cairo Institutes - Higher Institute of Computer Science and Information Systems

Expected Graduation: 2027

Certificates

- Python for Data Science and Machine Learning (Udemy)
- Certified Python Programmer (Udemy)
- Essential SQL Skills for Data Beginners
- Databases
- NLP and Deep Learning Specialization (Online Course)

Soft Skills

- Problem Solving
- Critical Thinking
- Communication
- Teamwork
- Adaptability
- Continuous Learning
- Time Management